

SETH FRIEND

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EDUCATION

University of Texas at Austin

Austin, TX

Electrical and Computer Engineering GPA: 3.79

Expected Graduation: May 2027

- Relevant Coursework: Software Design and Implementation II, Algorithms, Embedded Systems, Honors Probability, Honors Linear Systems/Signals, Digital Signal Processing

EXPERIENCES

Systems Engineering Intern - Hardware in Loop Simulation

Phoenix, AZ

Honeywell Aerospace (ValFac)

June – August 2026

- Engineered an automated Python data pipeline to parse SSI files and generate user-ready Excel workbooks, eliminating manual data formatting for Acceptance Test Procedures (ATP) documentation
- Developed an isolated model tester for simulation software in C++, significantly reducing simulation overhead
- Executed ATP for a Bombardier hardware-in-the-loop simulation bench, supporting validation of real avionics hardware

Electromechanical Lead

Austin, TX

Ivy Technology / Amazon Robotics

May – July 2025

- Led and coordinated teams of 7–10 to build 12 Cardinal automated work cells, delivering a week ahead of schedule
- Installed and wired 400+ automation-ready chutes in collaboration with other leads, integrating mechanical and electrical systems.
- Conducted quality assurance and troubleshooting on 20 Linux-based self-driving robots

Technical Assistant and Design Lead

Austin, TX

Texas Inventionworks | Design Team

Aug 2024 – Present

- Lead and support multidisciplinary student projects, providing guidance on embedded systems, fabrication, and design workflows

Machine Learning Researcher and Robotics Mentor

Austin, TX

Robot Interactive Intelligence Lab

Jan – Dec 2024

- Achieved 95% success training a robot arm in simulation using imitation learning (robomimic)
- Collaborated on a team of 3 to integrate a motion capture system as an input device for Telemoma
- Configured Ubuntu/ROS environments for lab computers, supporting ongoing robotics projects
- Provided individualized help to 20+ high school students in PyTorch, ROS, and kinematics

PROJECTS

Humanoid Robot | Software Development, Simulation, Control Systems, CAD

July 2025 – Present

- Designed a humanoid robot in Onshape with 5 DoF in each limb
- Developed Python motor bring-up scripts to configure BLDC actuators, validate encoder feedback, and automate initialization over CAN bus
- Created test script to interact with each motor over CAN bus
- Developed a physics-based simulation environment in Isaac Lab to test reinforcement learning strategies
- Taught robot to stand while undergoing translational forces in simulation

EXTRACURRICULAR EXPERIENCES

IMC Prosperity 4 (Team Mangoes) | Quantitative Researcher

2026

- Developed and backtested algorithmic trading strategies, validating simulated market data to achieve positive PnL and secure a top 7% international ranking

Engineers for a Sustainable World | Active Member

Aug 2023 – Present

Society of Hispanic Professional Engineers | Active Member

Aug 2023 – Present

TECHNICAL SKILLS

Languages: Python, C++, C, Java

Software Tools: Git, Linux, VS Code, Bash, Excel/OpenPyX

Libraries: Pandas, NumPy, Matplotlib, PyTorch